

TEST INVENTION DISCLOSURE

Synthetic document for testing an IPR / patentability audit application

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Invention Title	Adaptive Energy-Harvesting Smart Sensor Node
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1. Abstract

The invention is an adaptive wireless sensor node capable of harvesting small amounts of ambient energy from indoor light, temperature gradients, and mechanical vibration. The node dynamically selects an energy source and adjusts sensing, computation, and wireless transmission activity according to the available harvested energy. A local energy-management controller predicts short-term energy availability and schedules sensor measurements and communications so that the device can operate for extended periods without a conventional battery replacement.

2. Problem Statement

Conventional battery-powered IoT sensor nodes require periodic battery replacement or wired charging. This creates maintenance costs and limits deployment in locations where physical access is difficult. Existing energy-harvesting sensors may rely on a single energy source and may waste harvested energy when environmental conditions change. There is therefore a need for a sensor architecture that can combine multiple weak energy sources and intelligently adapt its operating schedule.

3. Detailed Description

The proposed node contains a photovoltaic harvesting element, a thermoelectric generator, and a piezoelectric vibration harvester. Each source is connected to an energy-harvesting interface containing rectification, voltage regulation, and source selection circuitry. A microcontroller measures source voltage, current, storage voltage, recent energy consumption, and environmental conditions.

A prediction module estimates the amount of energy likely to be available during a future operating interval. Based on this estimate, the controller selects an operating profile. During high-energy periods the node may increase sensing frequency and transmit larger data batches. During low-energy periods it reduces sampling frequency, delays non-critical processing, and transmits only priority measurements.

Energy is stored in a rechargeable storage element such as a supercapacitor or rechargeable micro-battery. The controller maintains a minimum reserve to prevent unexpected shutdown. The system may also combine energy from multiple harvesters when more than one source is simultaneously available.

4. Example Use Cases

- Indoor environmental monitoring in offices and industrial facilities.
- Structural monitoring of bridges, machinery, and rotating equipment.
- Remote temperature and vibration monitoring where battery replacement is difficult.
- Low-power agricultural or warehouse sensing installations.

5. Potential Novel Features

- Dynamic selection among multiple ambient energy sources.
- Prediction-driven scheduling of sensing and wireless transmission.
- Energy reserve management based on predicted future availability.
- Priority-aware data transmission during energy-constrained periods.
- Coordinated use of heterogeneous harvesters rather than independent source operation.

6. Example Claim Concepts

Claim concept 1: A wireless sensor node comprising multiple heterogeneous energy harvesting elements, an energy storage element, a sensing subsystem, a wireless communication subsystem, and a controller configured to select an energy source based on measured environmental energy availability.

Claim concept 2: The sensor node of claim concept 1, wherein the controller predicts future energy availability and modifies a sensing or communication schedule according to the prediction.

Claim concept 3: A method of operating an energy-harvesting sensor node comprising measuring harvested energy, estimating future energy availability, determining an energy budget, and dynamically scheduling sensing and wireless communication operations within the energy budget.

7. Prior Art / Known Approaches

Known approaches include battery-powered wireless sensor networks, photovoltaic-powered sensor nodes, thermoelectric generators, vibration-powered sensors, and adaptive duty cycling based on battery state. The applicant has not performed a comprehensive prior-art search for this synthetic test disclosure.

8. Development Status

Prototype architecture completed. A laboratory prototype has been conceptually specified but no commercial product has been released. This document is synthetic and is intended only for software testing.

IMPORTANT: This is fictional test data and is not a real patent disclosure, legal document, or statement of actual inventorship.